

Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	31 January 2026
Team ID	LTVIP2026TMIDS36402
Project Name	Heritage treasures: an in depth analysis of UNESCO world Heritage sites in tableau
Maximum Marks	4 Marks

HERITAGE TREASURE—ANALYTICS & VISUALIZATION WORKFLOW

(SQL, EXCEL, Tableau / Power BI)

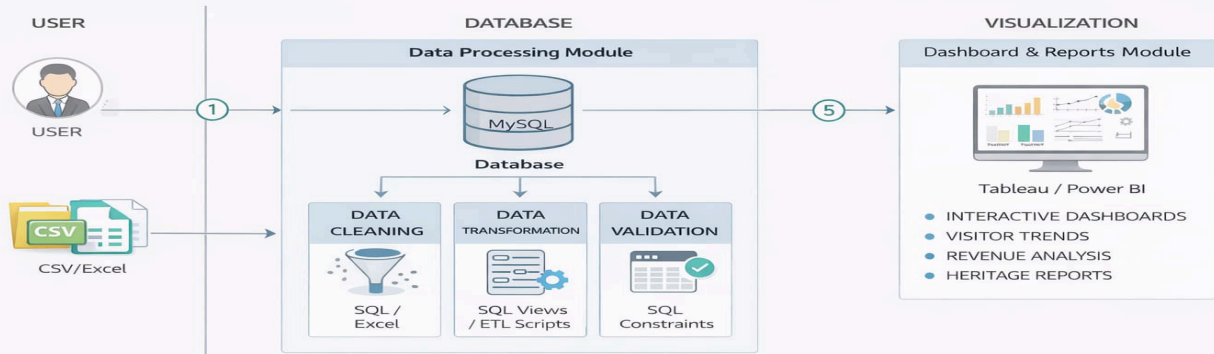


TABLE-1: Workflow Components & Technologies

S.No	Components	Technologies
1.	Data Source	CSV / Excel
2.	Data Storage	MySQL Database
3.	Data Processing	SQL / Excel / ETL Scripts
4.	Data Visualization	Tableau / Power Bi

Table-1 : Components & Technologies:

S.No	Component	Description	Technology
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1.	Data Source	Contains raw heritage site data such as visitor count, revenue, location, and category	CSV / Excel
2.	Data Storage	Stores organized heritage data securely for querying and analysis	MySQL
3.	Data Visualization	Builds interactive dashboards, charts, and maps for analysis	Tableau / Power BI
4.	Reporting	Presents insights like regional comparison, revenue trends, and visitor growth	Dashboard Reports
5.	Dashboard Interaction	Allows users to explore data dynamically using filters and slicers	Tableau Filters / Power BI Slicers
6.	Data Validation	Ensures accuracy, consistency, and reliability of data	Excel / SQL Constraints
7.	Data Processing	Cleans data, removes errors, handles missing values, and creates KPIs	SQL / Excel
8.	KPI Calculation	Computes total visitors, total revenue, growth %, and average visitors per site	SQL & Dashboard Calculated Fields

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	Used open-source tools for storing, processing, and visualizing heritage data.	MySQL, SQL, Excel, Tableau / Power BI
2.	Security Implementations	Implemented login authentication and restricted database access. Data validation applied.	RBAC, SQL Authentication, HTTPS
3.	Scalable Architecture	Followed 3-tier architecture (Database, Processing, Dashboard) for easy expansion.	MySQL, SQL, Tableau / Power BI

S.No	Characteristics	Description	Technology
4.	Availability	Dashboard accessible online for users anytime.	Tableau Server / Power BI Service
5.	Performance	Optimized queries and used filters for fast dashboard performance.	SQL Indexing, Data Extracts